# HOMEWORK 3

Relational algebra is used to let us see what operations in the relational model look like and how we can navigate through data between different relations. It can also be used to identify different kinds of constraints that are useful for restricting database content.

**Important notes:**

- Only use operators from the course book **(chapter 2.4 and 5.1-5.2)** or from the **lecture slides**.
- Splitting a long expression into several smaller ones using assignments is valid and sometimes recommended
- The book mentions both the "bag" and "set" model. In this homework, the "bag" model is assumed. See Chapter 5.1 in the book.

## REQUIRED TASKS (P)

1. If you received feedback to change regarding homework 2, or if you simply want to make adjustments of your own accord, implement these changes, clearly state what has been modified, and explain the reason behind each change.
2. List all your normalised schemas from the previous homework so it is clear what the relational algebra is based on.
3. Write a relational algebra expression for each of the following statements using your own schemas. For each expression, write an explanation for what is happening in the expression.
    a) A list of UserIDs of all the users that have not attended an event.
    b) A list of UserIDs of all the users who do not currently have a subscription.
    c) A list of names of all users that have liked more than 5 posts.
    d) A list of unique PostURLs containing an image or audio attachment that have been taken at the same place as the event with eventID = 1368. No PostURL may be listed more than once.
    e) A list of PostURLs of all the posts that have **both** Crypto **and** Studying as tags posted after the year 2023.

**Grading Criteria**:

- Is the feedback for homework 2, if such was given, implemented?

**NOTE: 1)** Homework **MUST** be handed in via Canvas **only** and in the correct assignment folder, in one file representing the entire homework. **2)** Your names, group number must be included in the document. **3)** Take into consideration this **exercise is accumulative** and builds upon your solutions. **4) All homeworks are based solely on the course case study.**

- Are all changes in task 1, if any were made, implemented correctly and explained?
- Are all the final schemas from homework 2 included in the list in task 2?
- Are all relational algebra expressions correct?
- Are the relational algebra expressions clear and readable, with detailed explanations?

# HOMEWORK 3 P+ (Constraints)

**Important requirement:** To be able to pass the P+ assignment, the P assignment has to pass.

**Notes:**

- P+ is graded more harshly than P
- Make sure you **read the book carefully.**

## TASKS (P+):

1. Write a relational algebra expression for each of the following statements using your own schemas. For each expression, **write an explanation for what is happening in the expression**.
   You may utilize your expression from a) in task b)
   *In this task, you wish to investigate whether attachments are more common in posts from events.*
   a. A list of all PostURLs for all posts that do not have attachments, followed by a column of zeros to represent that they do not have any attachments. To achieve this, you may use the table *Zero*, which is a table containing a single cell with the number 0. Make sure to give the column of zeros an appropriate attribute name.
   b. A list containing the average number of attachments on posts from events and from posts taken outside of events. A post is considered related to an event if it is taken at the same place and during the same time as an event.

2. Explain each of the following constraints and give at least 1 example for each from your database design.
   a. Referential integrity constraint
   b. Domain constraint
   c. Key constraint
   Each example must be presented as a relational algebra expression, which you must also explain and clarify.

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**Important notes:**

- For the example regarding domain constraints, you cannot use that attributes must be atomic or that each attribute only has one data type, since neither of these can be expressed as a relational algebra expression.

**Grading Criteria**:

- Are the explanations correct and thorough?
- Do the expressions follow the standard from the book?
- Are the relational algebra expressions presented in a readable manner?
- Is at least 1 correct example given per constraint?
- Is each example correctly expressed as a relational algebra expression?
- Is each relational algebra expression explained in words?
- Is the relational algebra expression correct?

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